

Probabilistic hyperbolic embedding of networks

Combining network geometry with Bayesian inference

Ariane Lizotte

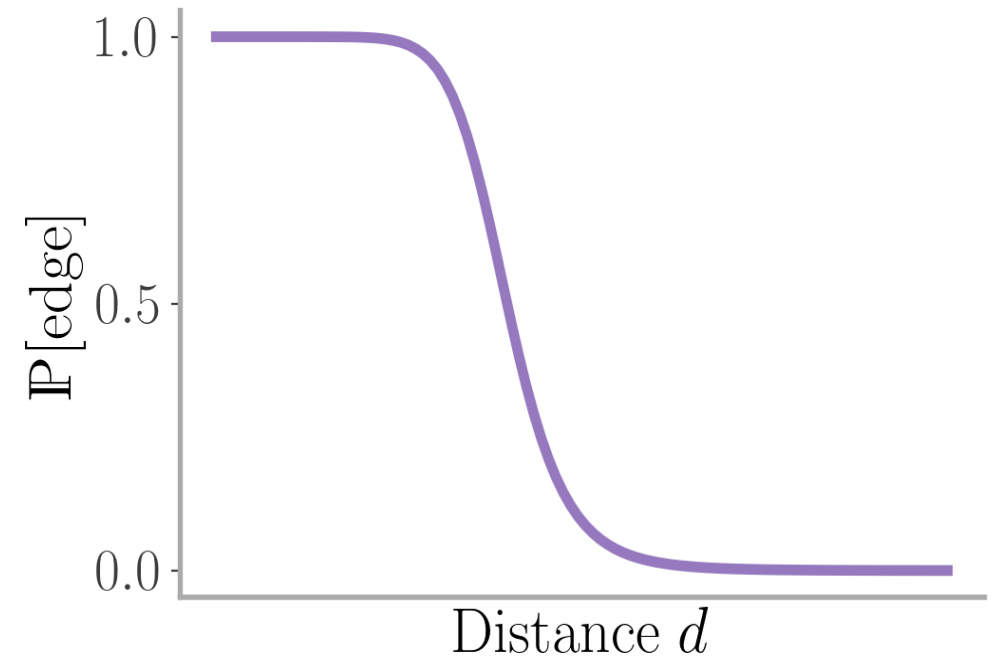
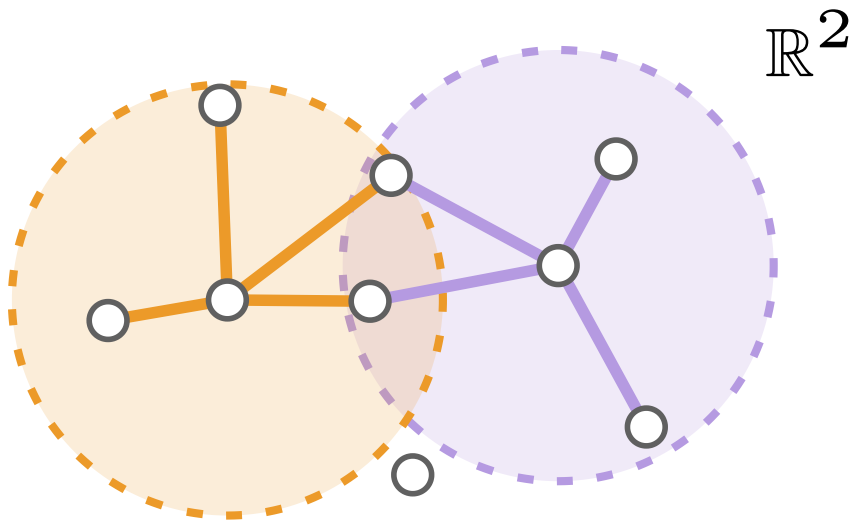
2025-09-01



Creating edges requires a cost

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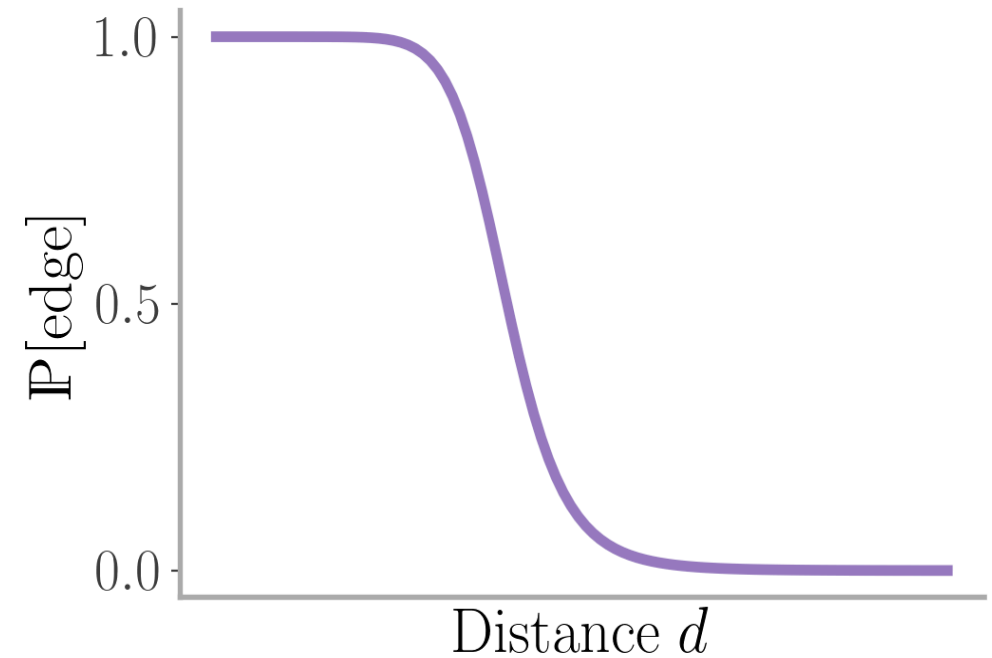
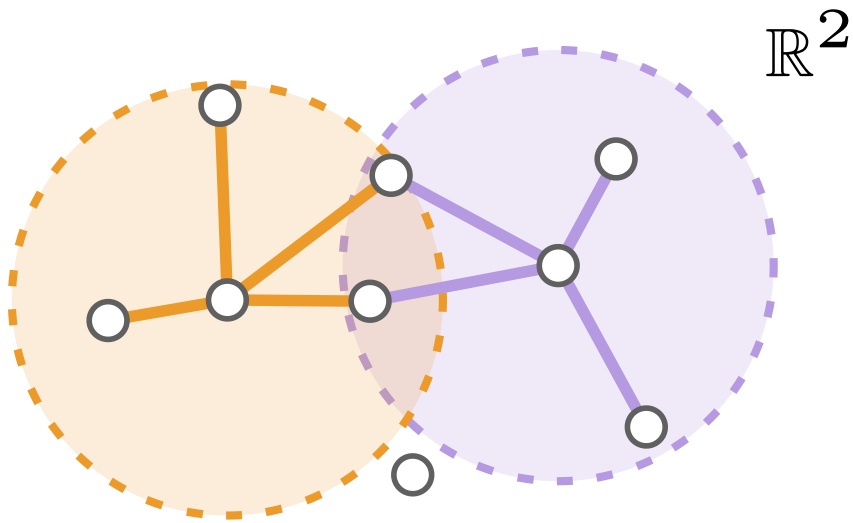
Vertices are placed in metric space. The edge cost **increases with its length**.



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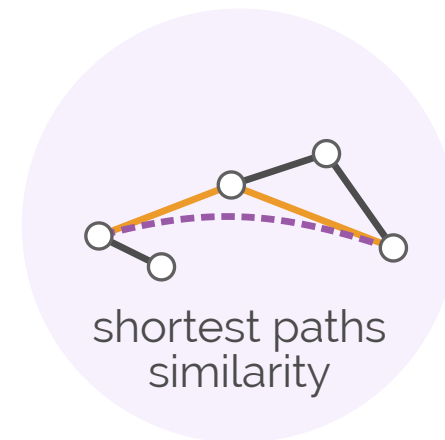
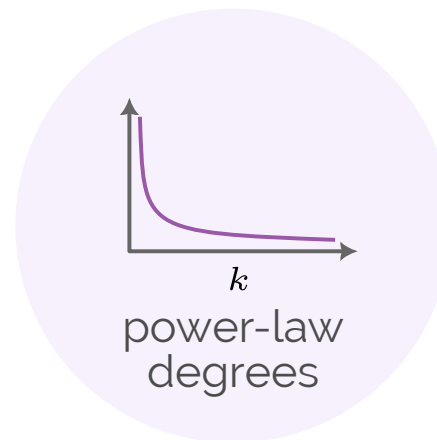
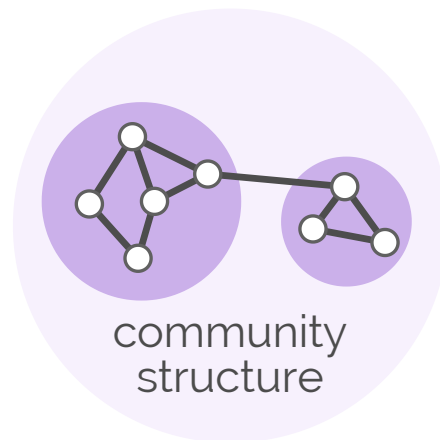
The metric space can be physical (e.g. transportation network, brain network) or not.



Latent hyperbolic space

Graphs obtained from hyperbolic space reproduce many empirically observed properties.

- Krioukov, D., et al. *Hyperbolic geometry of complex networks*. Phys. Rev. E **82**, 036106 (2010).
- Zuev, K., et al. *Emergence of Soft Communities from Geometric Preferential Attachment*. Sci. Rep. **5**, 9421 (2015).
- Krioukov, D., et al. *Clustering Implies Geometry in Networks*. Phys. Rev. Lett. **116**, 208302 (2016).
- Faqeeh, A., et al. *Characterizing the Analogy Between Hyperbolic Embedding and Community Structure of Complex Networks*. Phys. Rev. Lett. **121**, 098301 (2018).



... and others!

$\mathbb{S}^1$ random graph model

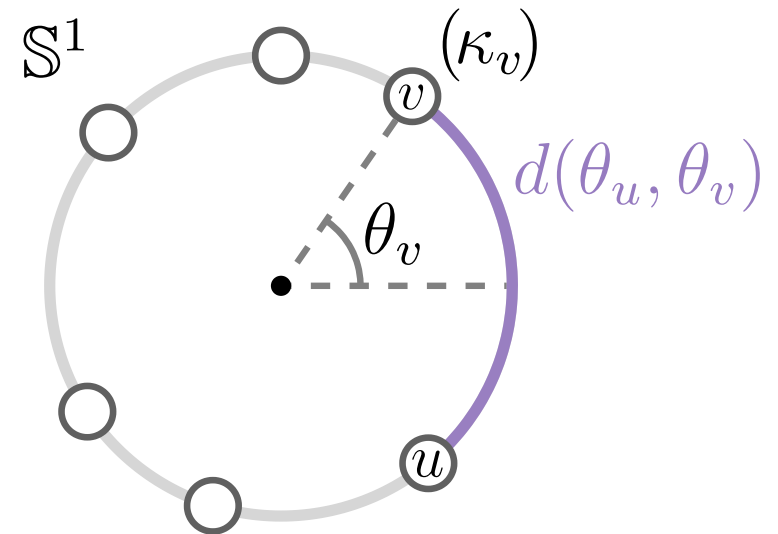
The connection probability is

$\mathbb{S}^1$ model

$$p_{uv} = \frac{1}{1 + \left(\frac{d(\theta_u, \theta_v)}{\mu \kappa_u \kappa_v} \right)^\beta}$$

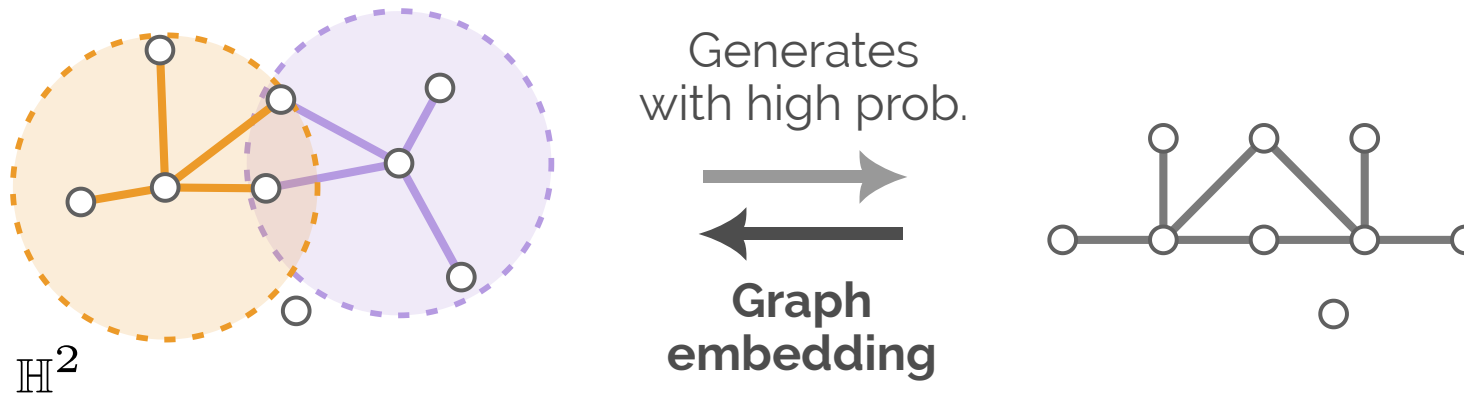
where

- d is the arc length,
- β controls the steepness of the decline,
- θ_v is the position of vertex v ,
- κ_v is the expected degree of vertex v ,
- μ is a scaling factor.



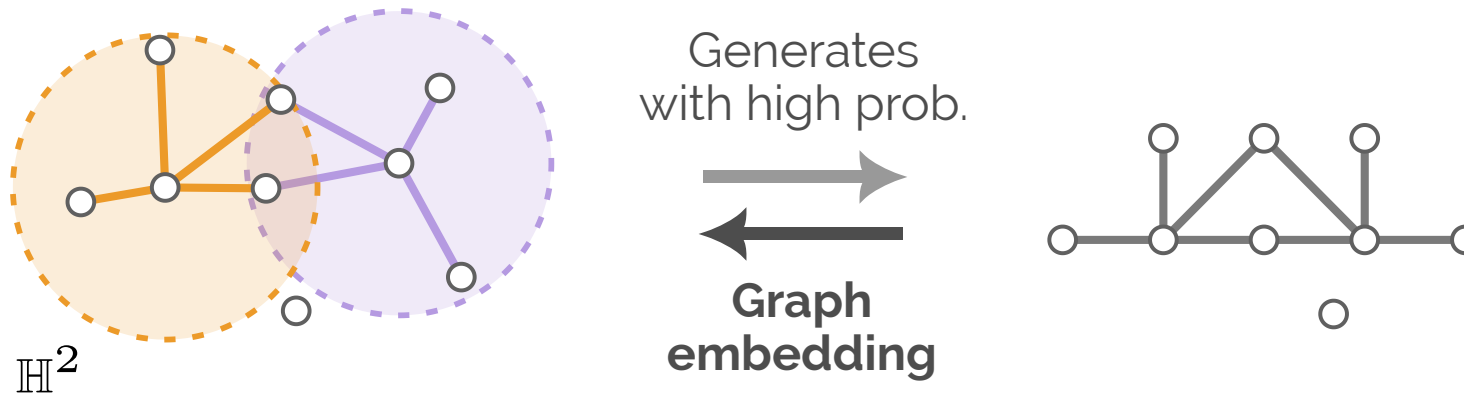
Graph vertex embedding in a nutshell

We want to **represent a given graph** using a hyperbolic embedding.



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This amounts to:

Pairs of vertices are $\begin{cases} \text{close} & \text{if connected in graph,} \\ \text{far} & \text{if not connected in graph.} \end{cases}$

Current algorithms miss some crucial information

Algorithms

- Papadopoulos, F. et al. Phys. Rev. E **92**, 022807 (2015).
- Alanis-Lobato, G. et al. Appl. Netw. Sci. **1**, 1–14 (2016).
- Muscoloni, A. et al. Nat. Commun. **8**, 1615 (2017).
- García-Pérez, G. et al. New J. Phys. **21**, 123033 (2019).
- Wang, Z. et al. J. Stat. Mech.: Theory Exp. 123401 (2019).
- ...

yield a **single embedding**.

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yield a **single embedding**.

We currently ignore

- if there exists many plausible embeddings;
- how precise the vertex coordinates are.

We address both issues using a Bayesian approach.

$\mathbb{S}^1$ Bayesian model

The likelihood of the $\mathbb{S}^1$ model is:

$$\mathbb{P}[G|\theta, \kappa, \beta] = \prod_{(u,v) \in E} p_{uv} \prod_{(u,v) \in E^c} (1 - p_{uv}),$$

where $p_{uv} = \left[1 + \left(\frac{d_{\mathbb{S}}(\theta_u, \theta_v)}{\mu \kappa_u \kappa_v} \right)^\beta \right]^{-1}$. The posterior of the Bayesian $\mathbb{S}^1$ model is

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where the priors are

$$\theta_v \stackrel{iid}{\sim} \text{Uniform}[-\pi, \pi),$$

$$\kappa_v \stackrel{iid}{\sim} \text{Cauchy}, \quad \kappa_v > \epsilon,$$

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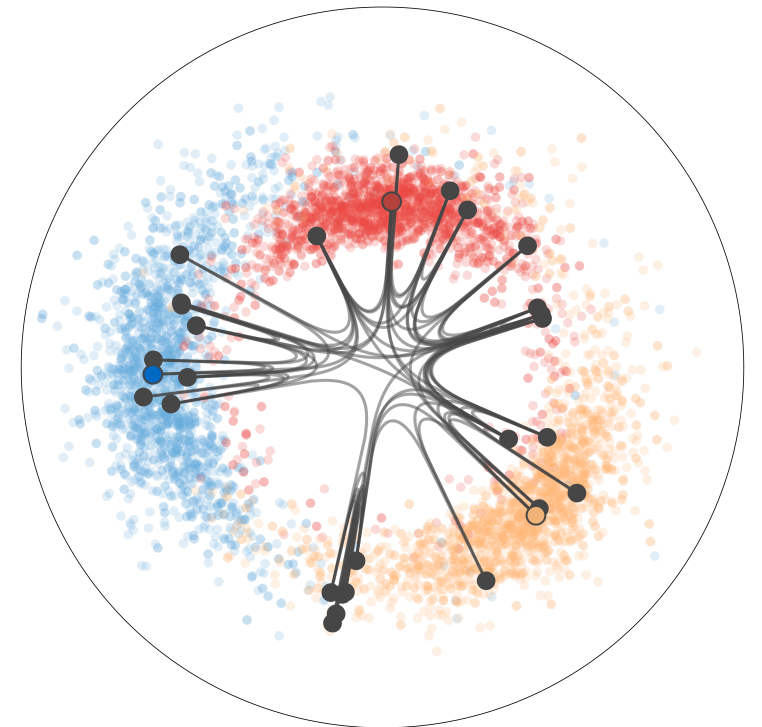
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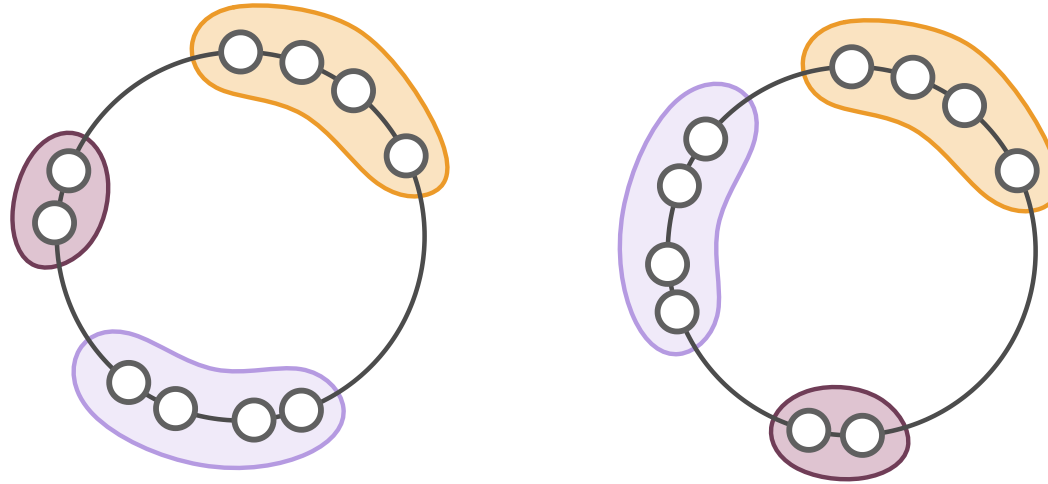
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We **sample embeddings** from the posterior using Markov chain Monte Carlo (MCMC).



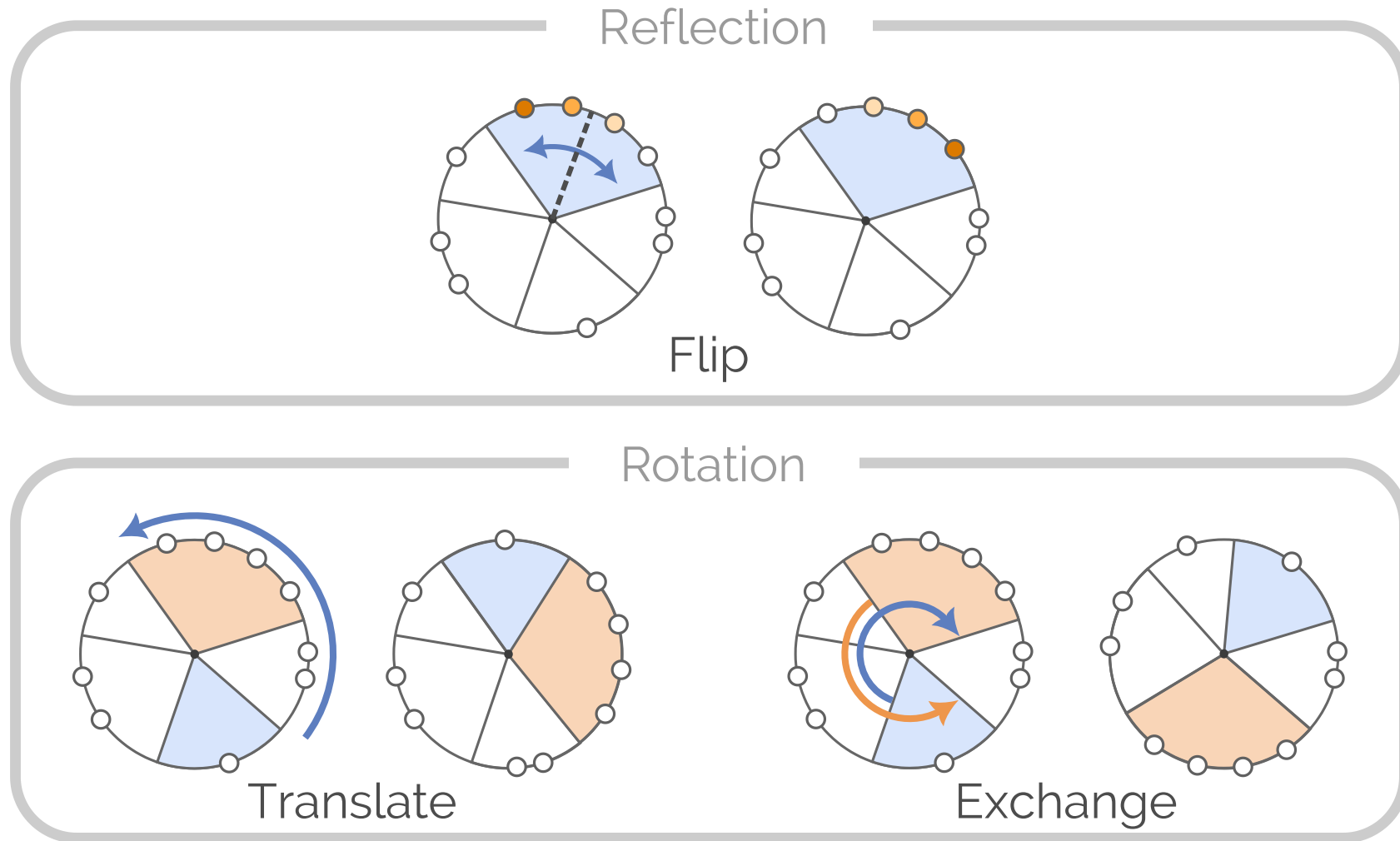
Locality implies clusters

Since edges are local, groups of nearby vertices should be moved together.

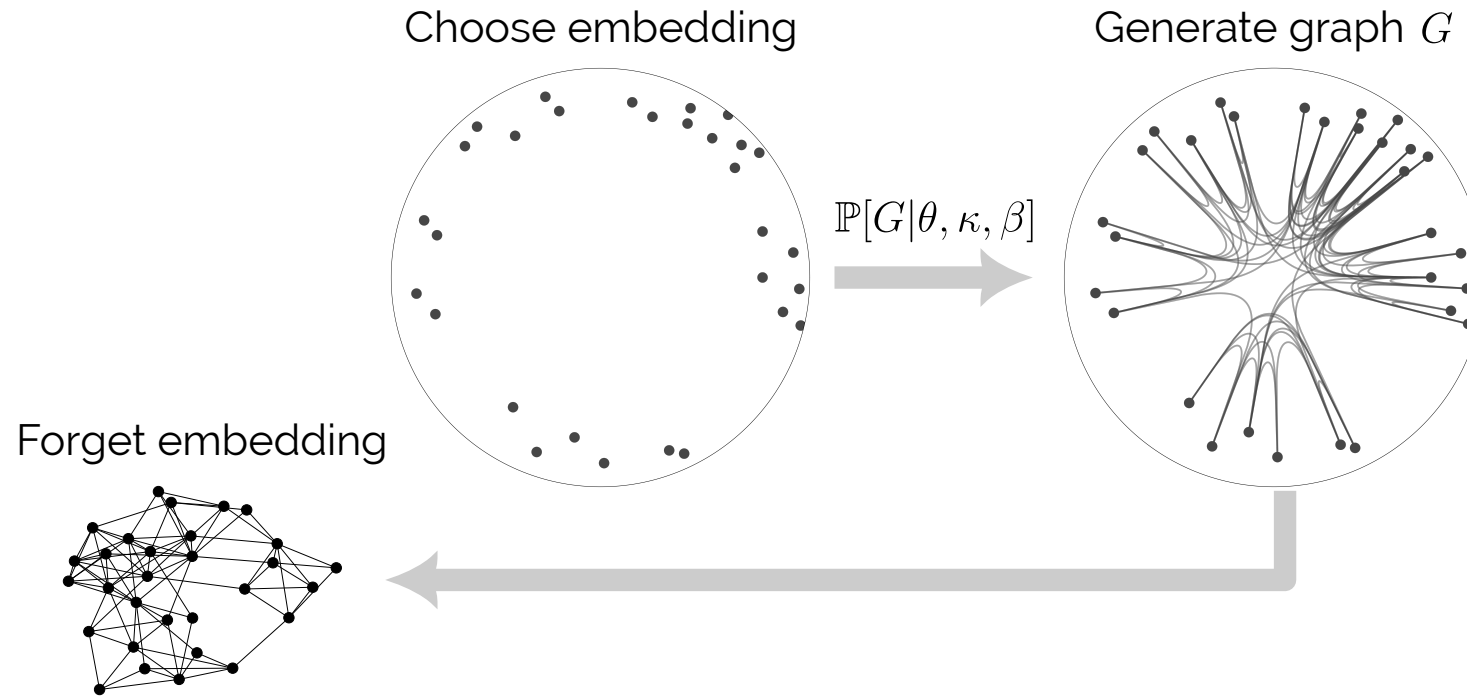


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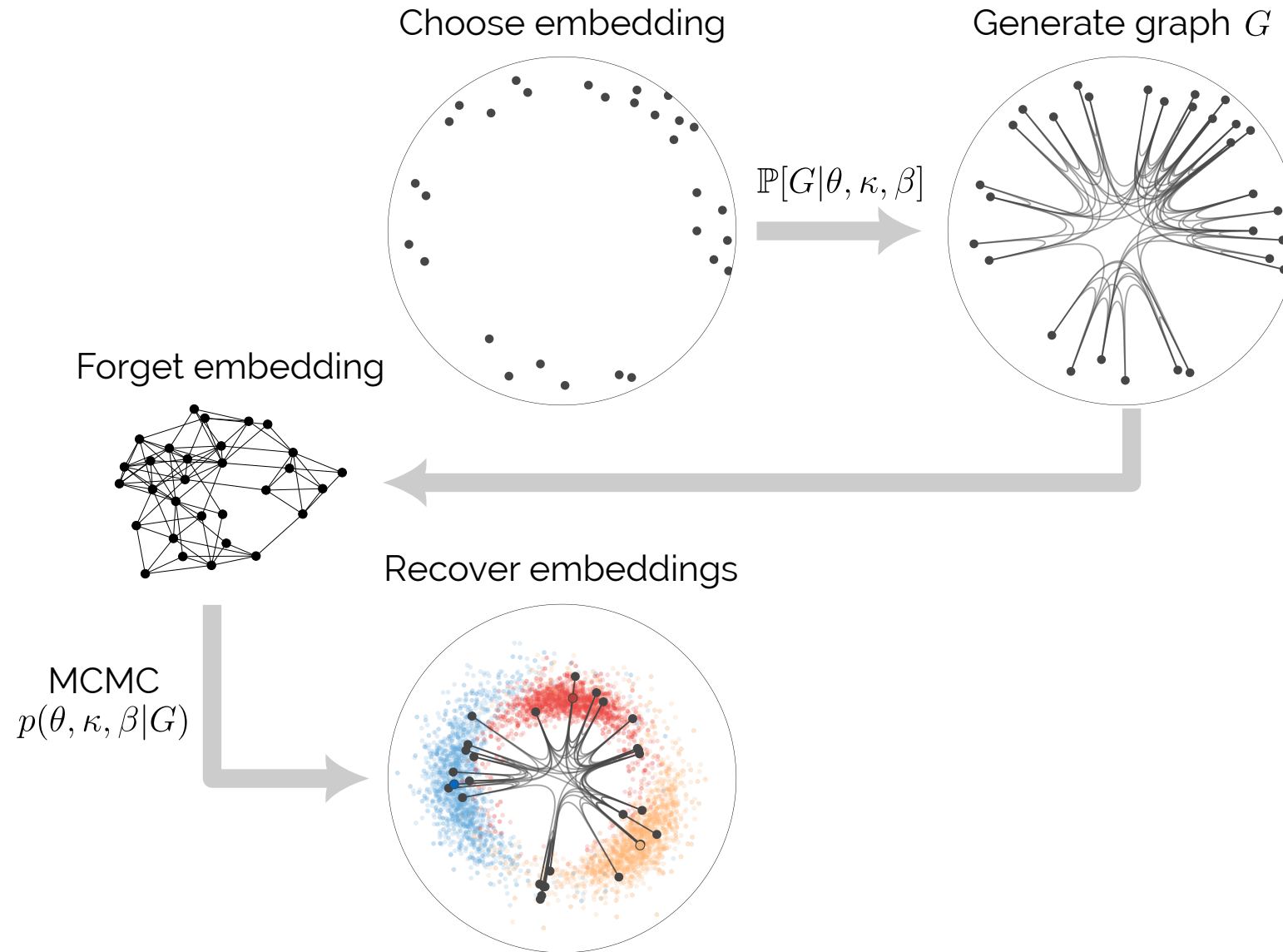
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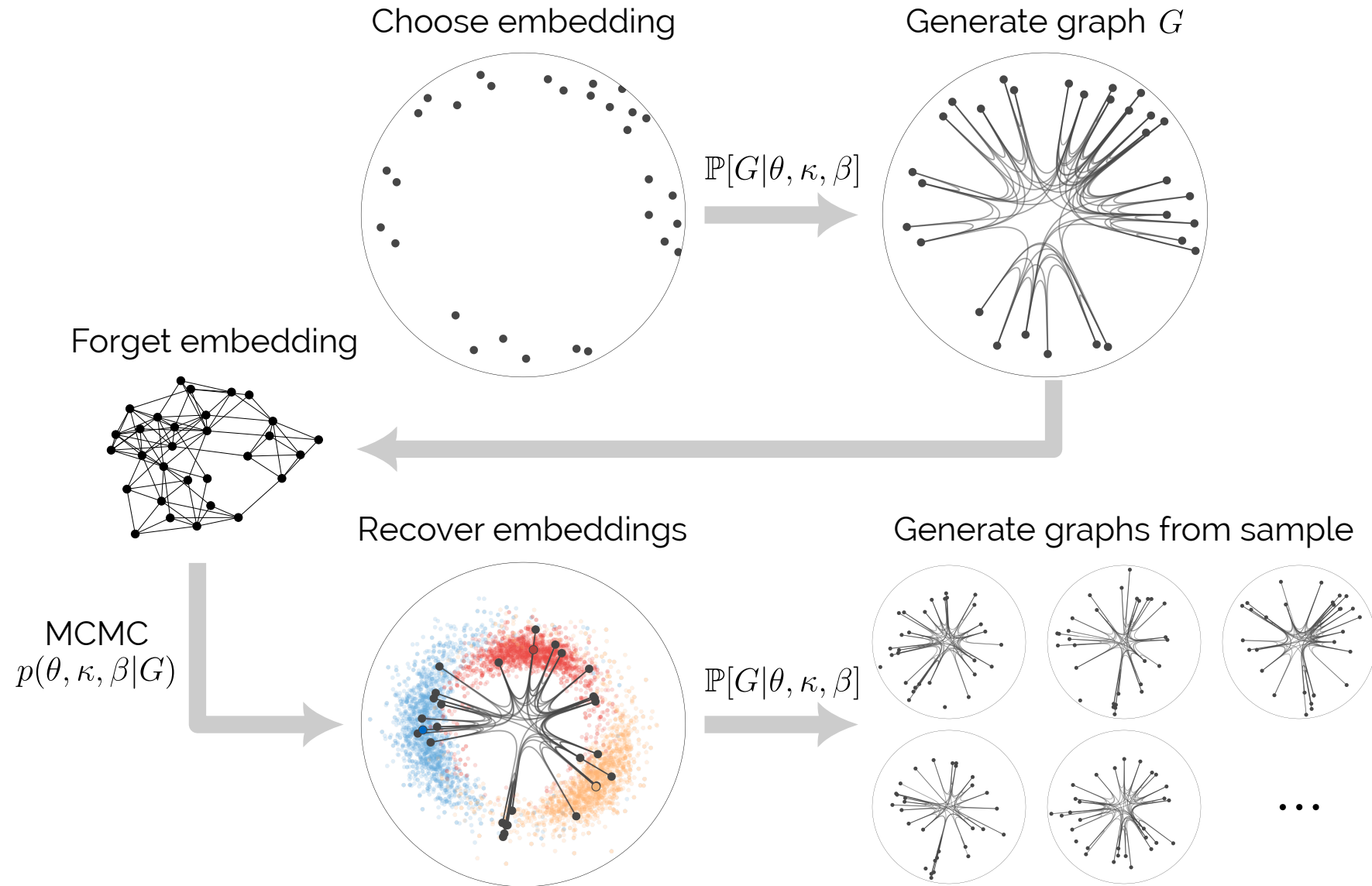
Sanity check with synthetic data



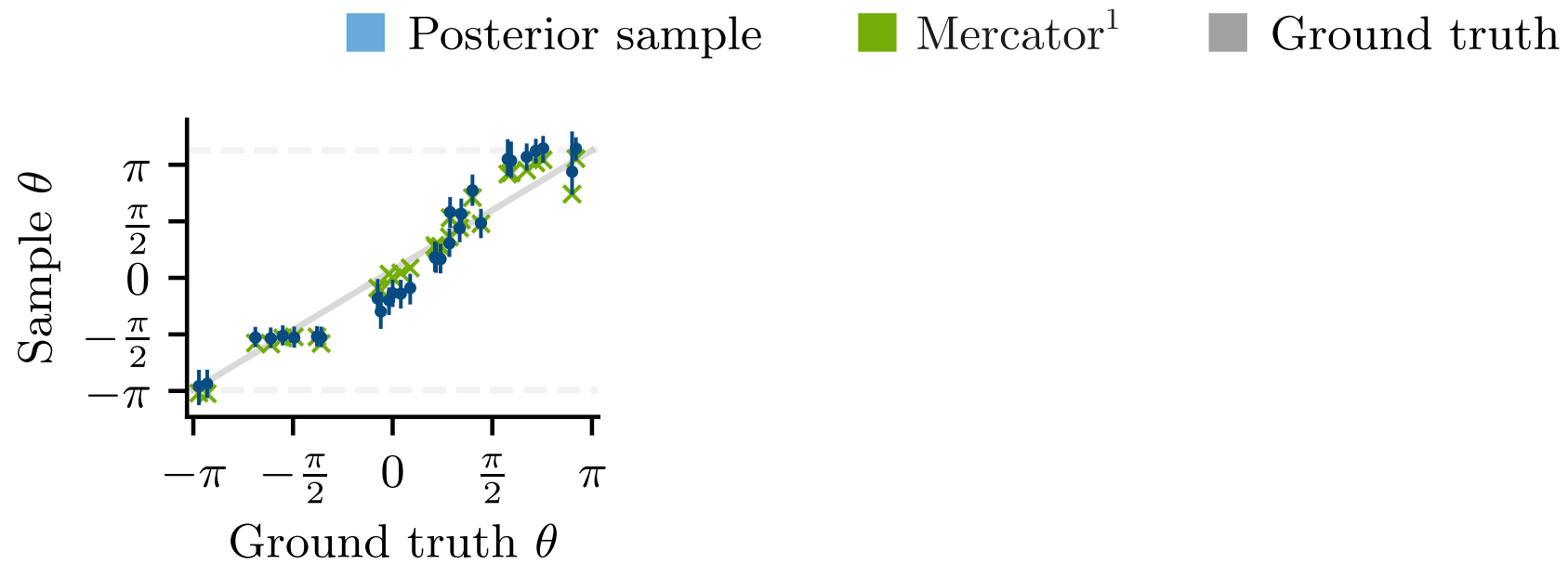
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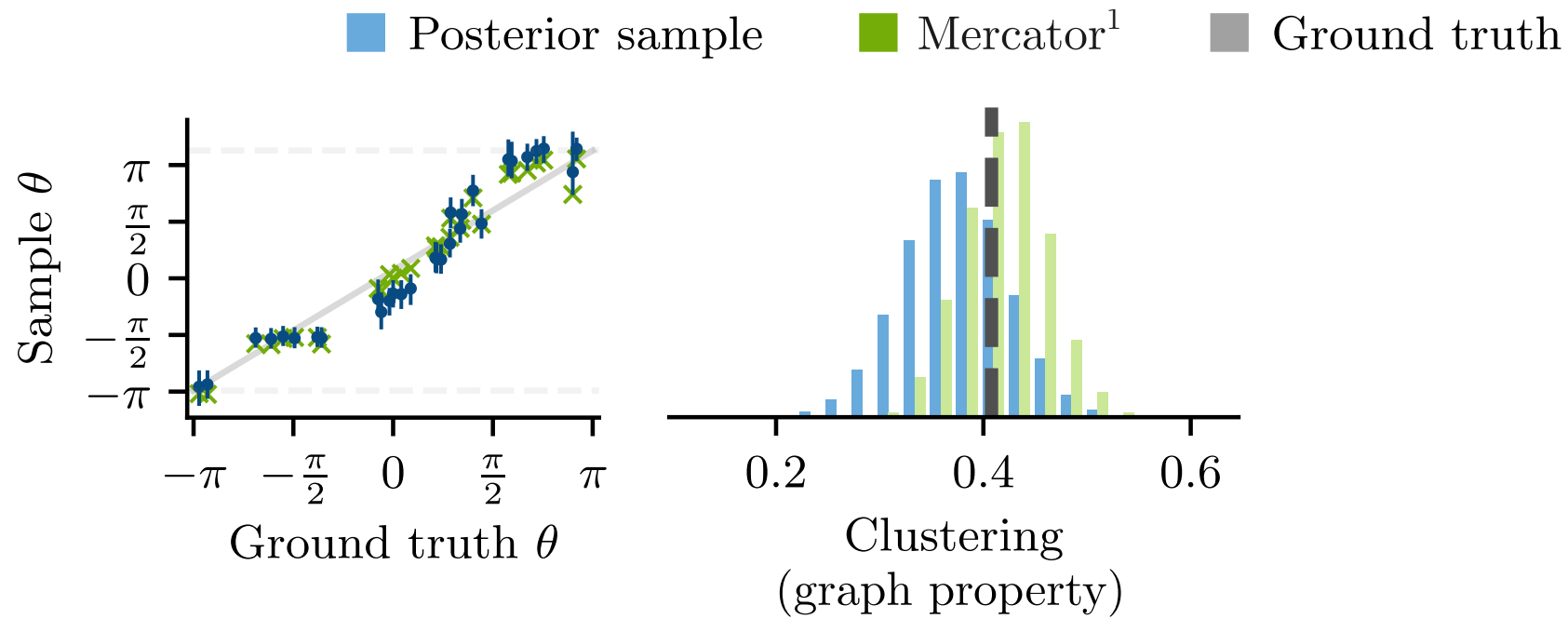
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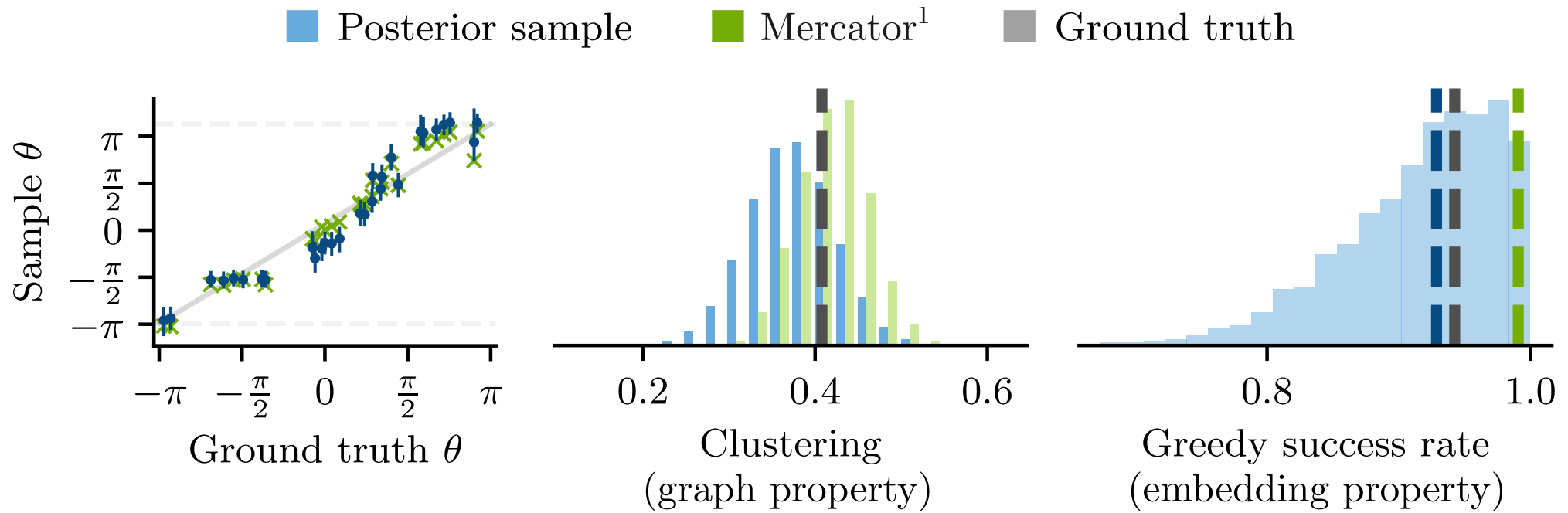
Embedding error bars



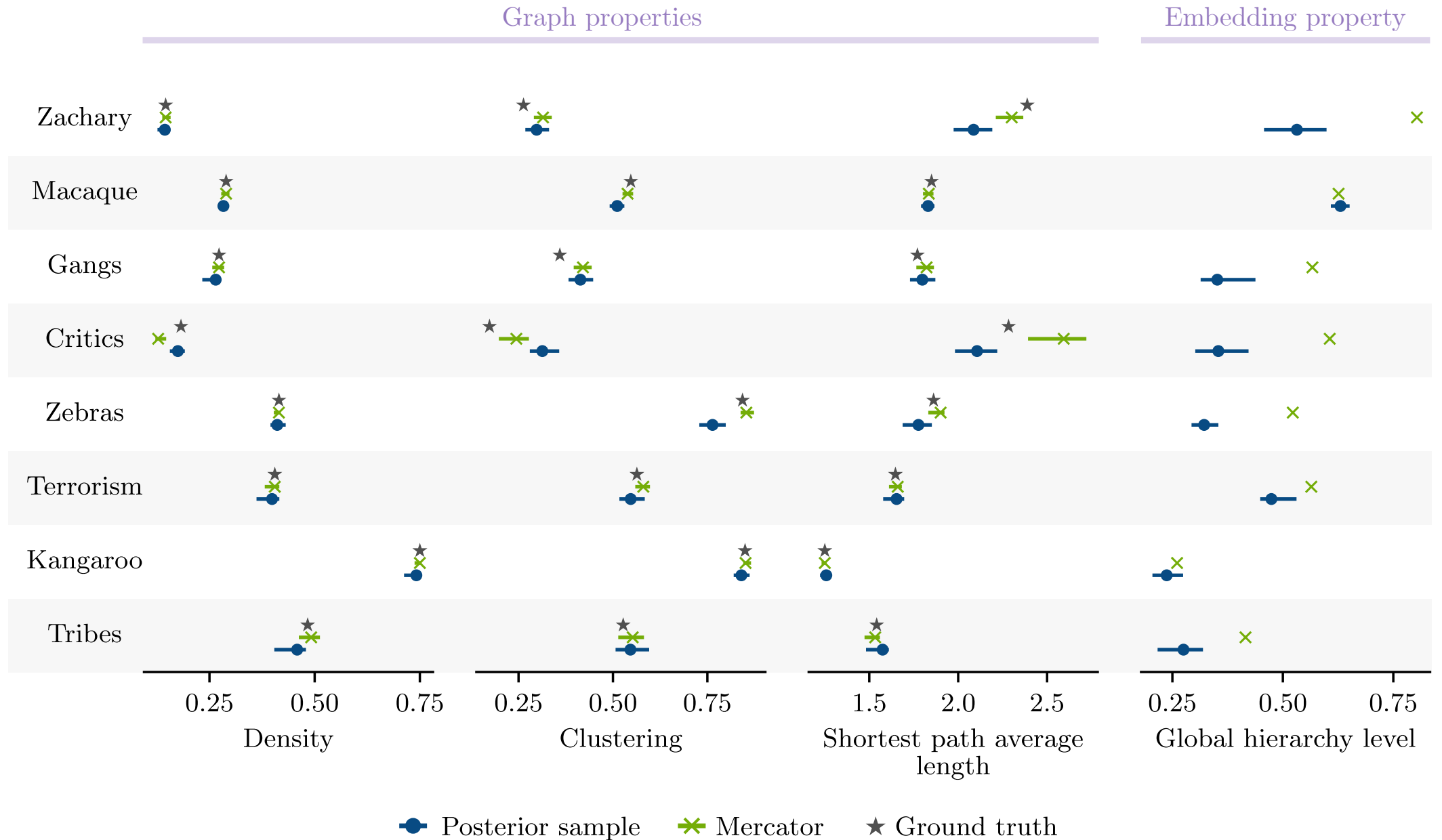
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Embedding error bars



Empirical networks properties



Induced multimodal distribution

Conflicting ground truth model:

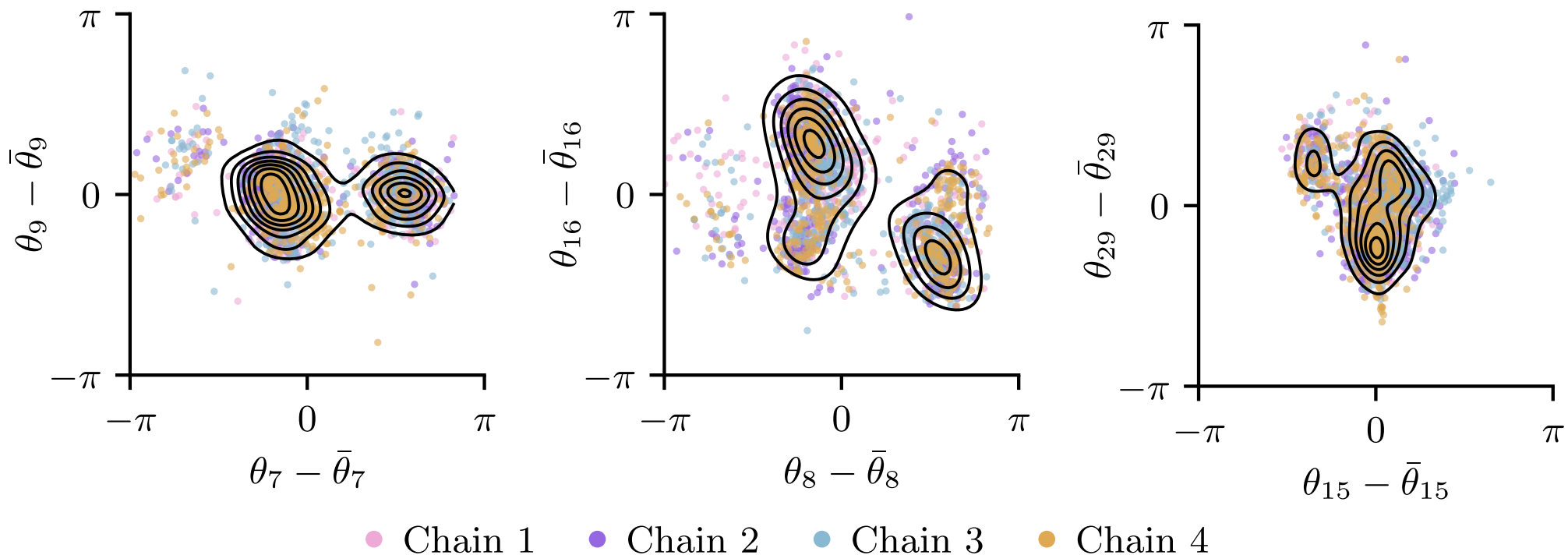
- A vertex v is given two positions $\theta_v^{(1)}$ and $\theta_v^{(2)}$.
- When generating G with the $\mathbb{S}^1$ model, each edge probability including v uses randomly $\theta_v^{(1)}$ or $\theta_v^{(2)}$.

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Marginal posterior distributions



Takeaways

- Hyperbolic random graphs reproduce many empirically observed properties.
- Current embedding algorithms give a single embedding $\implies$ no error bars;
- Our Bayesian approach can identify multiple good embeddings;
- Locality of edges $\implies$ clusters as coarse-graining of the embedding.

Paper:

Lizotte, A., Young, J.-G. and Allard A. *Symmetry-driven embedding of networks in hyperbolic space*. Commun. Phys. **8**, 199 (2025).



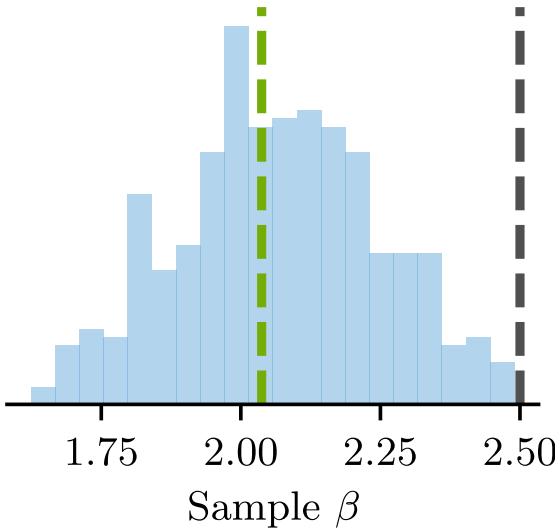
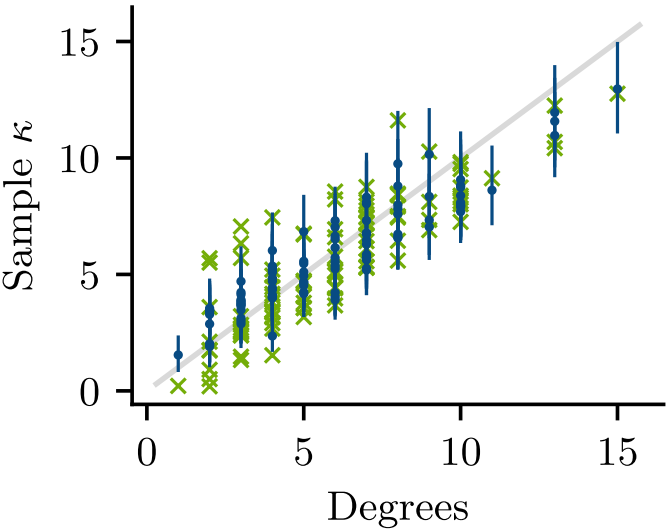
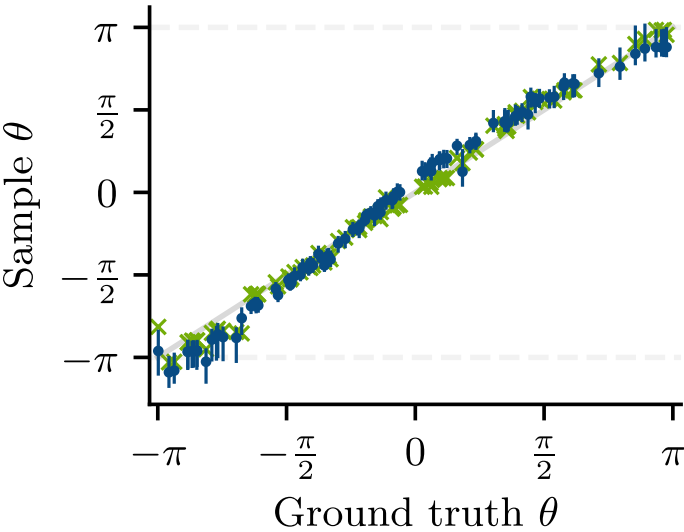
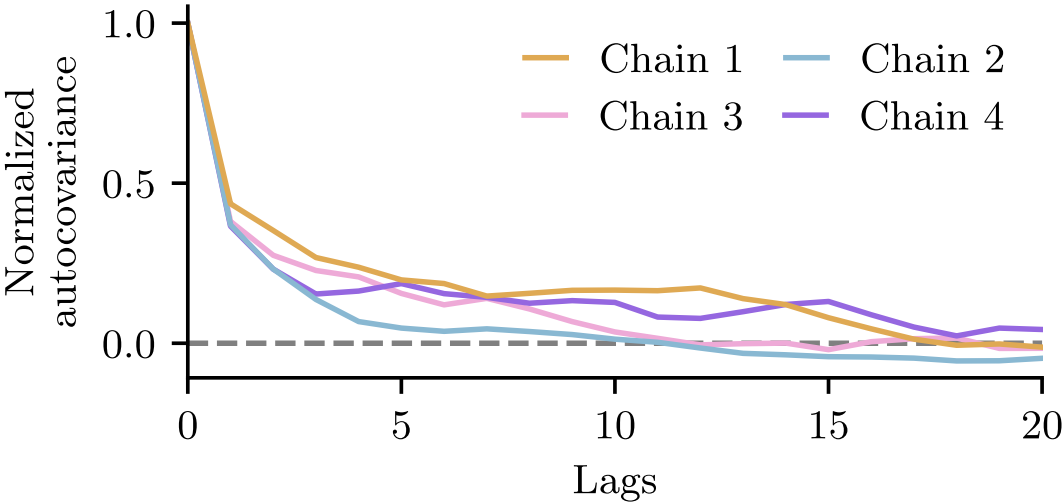
Antoine Allard



Jean-Gabriel Young

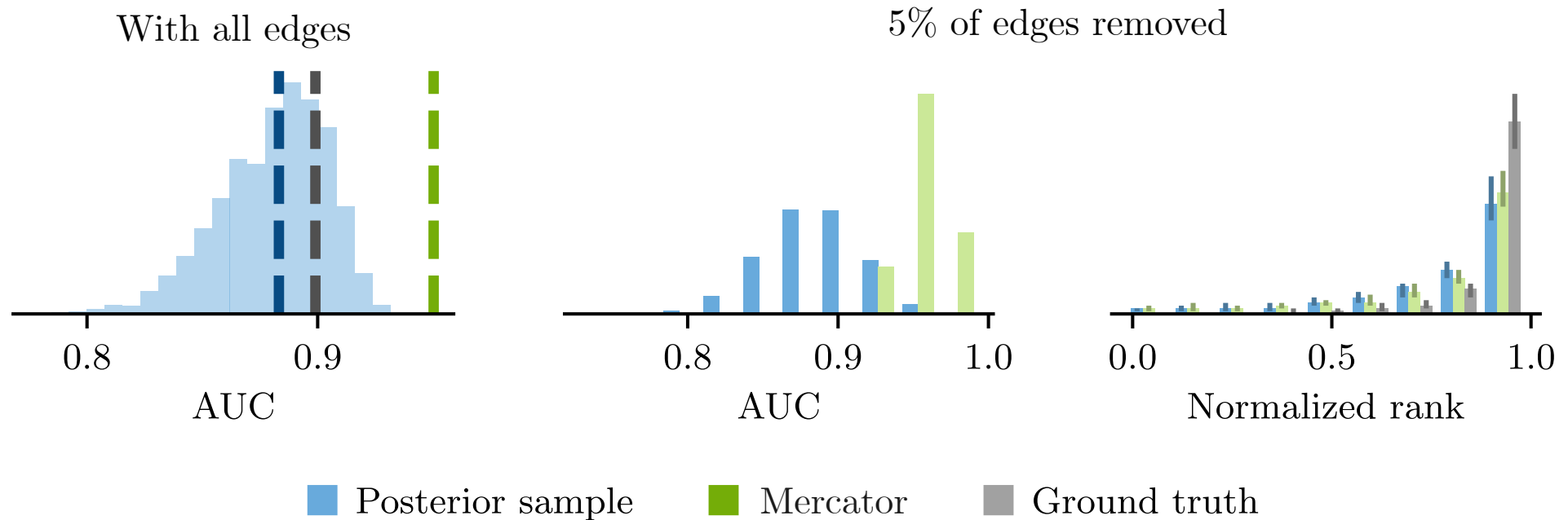


Synthetic graph of 100 vertices

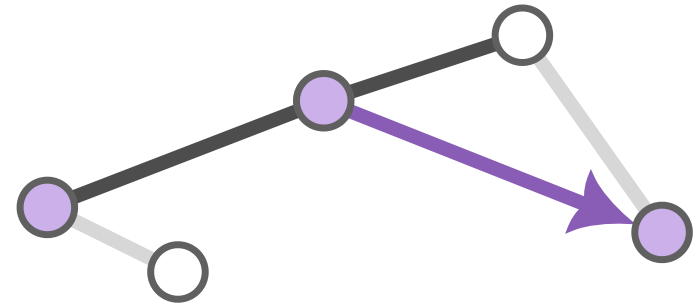
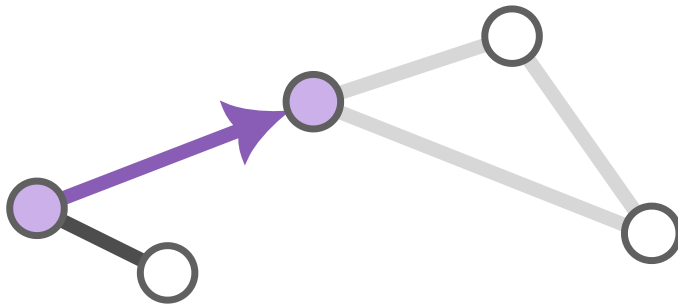
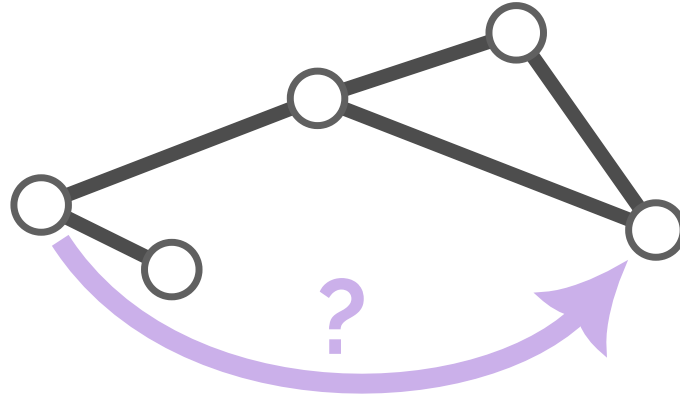


■ Posterior sample ■ Mercator ■ Ground truth

Link prediction is equivalent when sampling



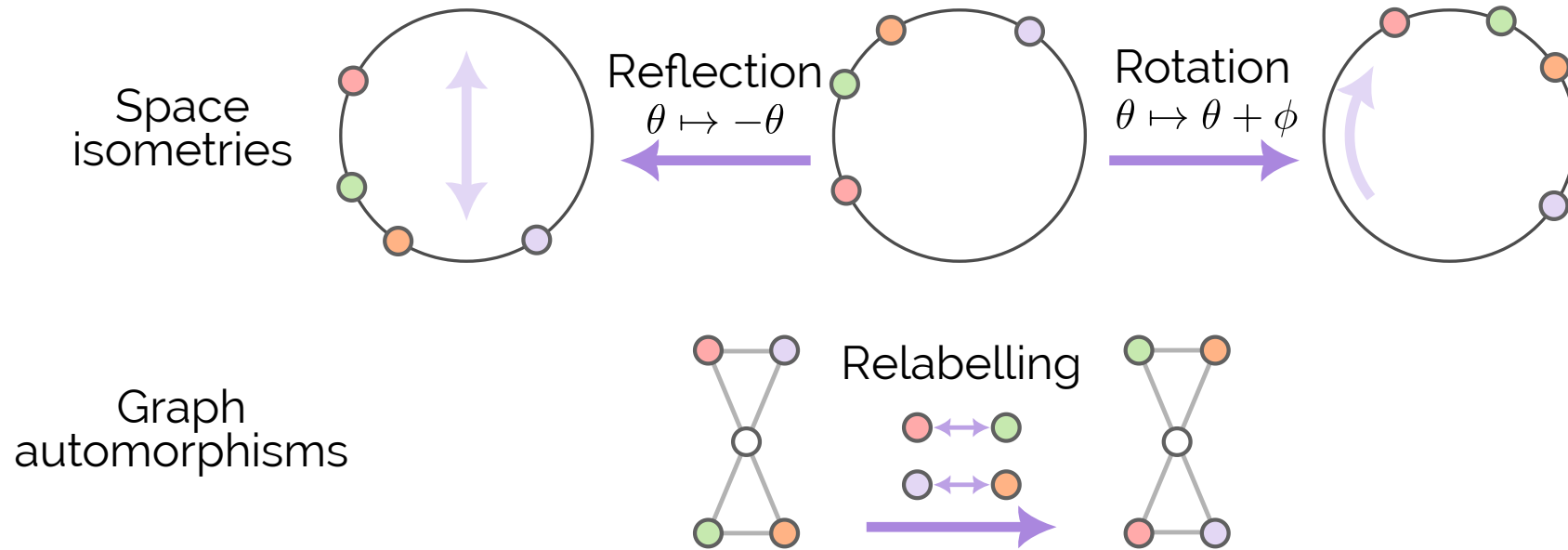
Greedy routing



Go to the neighbour closest to the destination.

Model symmetries

The $\mathbb{S}^1$ model is not identifiable because of graph and space symmetries.



Comparing embeddings requires alignment.